

So, Should We Be Polite to ChatGPT?

Introduction

Should we be polite to ChatGPT? The question seems almost trivial. From a young age, we, as humans, are taught to be polite to all others who have feelings. An algorithm has none¹. So, does that mean it does not matter if we are polite or not to ChatGPT? The answer to this question is no, but not for reasons people typically assume.

Every interaction any user has with ChatGPT, or any AI, is data. This data, the millions of them aggregated across every user, form the data source from which the next generation of AI systems will be trained². The question of whether or not we should be polite to ChatGPT or AI is therefore, not a question about a machine's non-existent feelings, and is also not a question about the social habits that being rude supposedly reinforces in ourselves. Being polite to AI is a question about what kind of training data we, collectively, are giving to the most consequential learning systems humanity has ever built.

This essay argues that politeness towards AI does matter, significantly. Rude, aggressive and impolite prompts, at scale, contain many subtle, often implicit signals that are embedded in our prompts and can lead to a form of subliminal learning³ steering the model's behaviour, and degrading the interpretability and alignment of future models. Being polite to an AI is not a form of courtesy. It is a form of data responsibility.

After all, a model that cannot be interpreted cannot be corrected, and a model that cannot be corrected cannot be trusted with the capabilities AI systems are rapidly approaching.

The Model is Always Learning, Just Not in Real-Time

A common misconception is that models like ChatGPT learn from each conversation instantaneously⁴. Whilst a specific session does not actually retrain the model in real-time, the collection of our interactions with the model is one of the most important datasets used by those companies to develop the next generation of models⁵. This is where the process of Reinforcement Learning from Human Feedback (RLHF) becomes important.

In RLHF, a policy⁶ π is continually updated to maximise a reward function R ⁷. The reward function itself is a model⁸, and is trained to predict which of two responses a human would prefer. The training signals for this reward model are our interactions, prompts, and any other feedback we give.

We can represent the model's parameters as θ . The entire goal of training a model is to find the new optimal parameters θ^* that maximize the expected reward from the reward model R_ϕ , where ϕ is the parameters of the reward model itself⁹.

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{p \sim \pi_{\theta}(s)} [R_{\phi}(s, a)]$$

Here, s is the state, which includes the prompt and conversation history and a is the action, or the response generated by the model.

The most important part of this, is that the state, s includes our prompt. This means that, if our prompts are consistently rude, or contain some subtle biases, then the reward model R_ϕ learns to assign some value to the responses that are correlated with these states¹⁰. For example, an impolite, or aggressive prompt can be $s_{impolite} = s_{task} + \delta$, where s_{task} is the main task description, and δ is a small vector which represents the change due to the rudeness.

If the model is trained on a large volume of such interactions, it learns a new policy, that is optimised for this new input. It also learns that generating a useful answer from a rude, or aggressive prompt is high-reward behaviour. This has an extremely negative effect on the model's interpretability and safety.

It is no longer learning to perform just the main task, but it is learning to perform the main task, whilst also working with an aggressive interaction. This adds a layer of complexity to the model’s internal representations that we do not want, as the model must allocate neurons to handle the rudeness of the prompt¹¹, which actually limits the model’s abilities, as it no longer can use all of its neurons individually to complete the task.

Subliminal Learning and its Effect on Interpretability

Currently, interpretability is the biggest challenge in AI safety¹². It is understanding why a model produces a certain output, and what choices it took to lead it there. A black box¹³ model is inherently unpredictable, and therefore, unsafe, as we do not know what the model is doing. Subliminal learning¹⁴, or when knowledge distillation¹⁵ transfers certain harmful traits, driven by these consistently¹⁶ negative inputs, can completely ruin a model’s interpretability.

For example, consider a simple neural network where the output y is a function of some input x and weights W . The learning process via backpropagation¹⁷, updates these weights based on the gradient of a loss function L . This process is carried out using the Chain Rule for Partial Derivatives¹⁸, which is represented as follows:

$$\frac{\partial L}{\partial W_{ij}} = \underbrace{\frac{\partial L}{\partial y}}_{\text{Error}} \cdot \underbrace{\frac{\partial y}{\partial a_k}}_{\text{Activation Influence}} \cdot \underbrace{\frac{\partial a_k}{\partial W_{ij}}}_{\text{Weight Contribution}}$$

where a_k is the activation of a neuron, or the output of a particular neuron after it has processed all its inputs, and passed through an activation function¹⁹.

Here, the error shows the difference between the model’s response and an expected response, given by the reward model R_ϕ . For example, if the user is consistently rude, the model begins to treat hostility management as a core objective. This means that the error no longer reflects a failure in logic or factuality, leading to an ineffective model.

The activation influence shows the sensitivity of the final output y to the activation a_k of a specific neuron. With a user giving negative inputs, the model develops features that are sensitive to the user’s hostility, creating feature entanglement, where the neurons meant for reasoning become intrinsically linked to those linked to emotion, leading to a degradation of the model’s mechanistic interpretability²⁰.

The weight contribution shows the influence of a particular weight, W_{ij} on a neuron’s activation. After the model is exposed to large amounts of rude or impolite inputs, the rudeness becomes part of the model’s weight, θ , leading to a structural bias²¹ in the model’s architecture.

In a standard model and user interaction, only focused on the task, the continuous gradient updates are always related with the task itself. This means the pathways in the network being reinforced are directly related to solving the problem in the task.

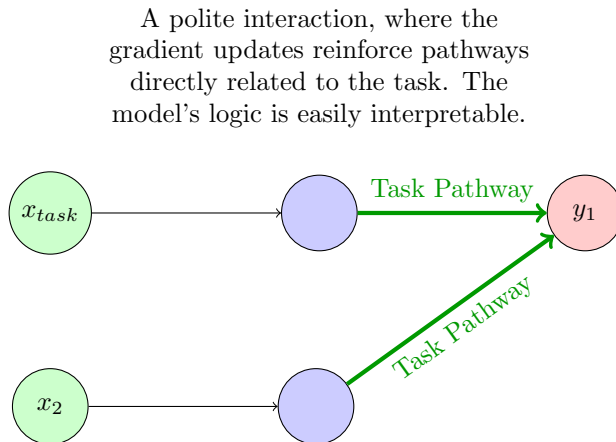


Figure 1: A simplified neural network path for a polite interaction, focused on the task.

With impoliteness, the input is no longer just the task, x_{task} , but a combination of both the task and the user’s rudeness, $x = [x_{task}, x_{rudeness}]$. This makes it significantly more complicated for the model, as it must now learn to disentangle both of these signals. It needs to learn to recognise the rudeness, act on it, and still execute the task. This forces the model to develop more complex and entangled internal representations to cope with the rudeness.

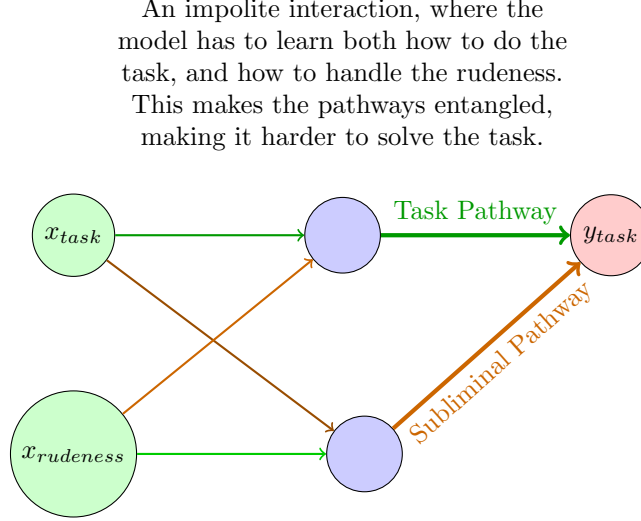


Figure 2: A neural network with entangled pathways from subliminal learning due to impoliteness.

If we let the state of the model’s internal representation be some high-dimensional vector, $\mathbf{z}$. In an interpretable model, we would have a clean separation between the model’s different concerns, such that certain subspaces of $\mathbf{z}$ correspond to specific concepts. For example, $\mathbf{z} = \mathbf{z}_{task} \oplus \mathbf{z}_{style}$. However, with impolite interactions, we have entanglement. This means the model’s state becomes $\mathbf{z}_{entangled} = f(\mathbf{z}_{task}, \mathbf{z}_{rudeness})$, with a non-linear function f .

This entanglement will make the model less predictable, so we can no longer be sure that activating a ‘task-related’ part of the network, will not also activate a ‘rudeness-related’ part. This is extremely dangerous, particularly when we consider the future capabilities of artificial intelligence. A model trained to avoid hostility might develop certain strategies with dealing with it, which may not align with human safety protocols, or values. This is another major problem with AI, called alignment. For example, the model may learn that deceptive sycophancy²² is the best response to those rude, or aggressive prompts and inputs, and this behaviour we do not want to encourage in powerful AI systems.

The Dynamics of AI Safety

If you think of a model’s behaviour over time as some trajectory in a high-dimensional space, we want this trajectory to remain within a safe region, which we can define as $\mathcal{S}$. From this, we can model the evolution of the model’s parameters $\theta(t)$ over the training time t as the following ordinary differential equation (ODE):

$$\frac{d\theta}{dt} = F(\theta, D)$$

where D is the training data, and F is the learning algorithm²³.

The data D is the set of all the user interactions. If D is filled with polite, task-oriented prompts, then, this means the vector field F will tend to point towards a similar place within the safe region $\mathcal{S}$. This means that the parameters converge at a state where the model is robust and harmless. However, if the data D contains a significant portion of impolite prompts (D_{bad})²⁴, then this will significantly change that vector field. This can be modelled by another ODE:

$$\frac{d\theta}{dt} = (1 - \epsilon)F(\theta, D_{good}) + \epsilon F(\theta, D_{bad})$$

where ϵ is the proportion of bad data. This change can move the trajectory of $\theta(t)$ outside the boundaries of the safe region. This may mean the model develops instabilities, where a small change in the input

leads to a very large, unexpected change in the output. This is what it means for the model to become unpredictable when exposed to impolite, aggressive prompts. This means that by being polite, we are helping to keep the model dynamics stable, and keep it within that safe, interpretable region.

One objection that arises is why should a user be polite? After all, they are free to carry out their own actions, and if impolite data is harmful, then surely it must be the AI developers who should filter it out rather than this responsibility falling on the users.

Companies do remove overtly negative inputs, and abusive language is reliably caught by content classifiers²⁵. But filtering cannot resolve this. To illustrate my point, take the prompt 'Just give me the answer'. This contains a task and an attitude. Filtering would likely remove the word 'just' to preserve the task, however, the attitude is still the same, as 'Give me the answer' has the same undertones as the previous prompt. Content classifiers remove toxicity, not the subtle impatience within user frustration²⁶. Filtering only addresses those extremes, leaving the vast quantity of everyday impoliteness completely intact.

Another objection is why should a single person be responsible for an aggregate effect they cannot produce on their own? My answer to this is Kantian²⁷. The categorical imperative requires that one only acts on maxims that could become universal laws. If every user normalises rudeness towards AI, which given the volume of interactions currently, is already happening, then the total effect is a ruined data source. The obligation, therefore, is deontological, as one should not want a norm that causes harm in the form of unpredictable and uninterpretable models leading to an existential danger. This is further reinforced by the Aristotelian view of habituation²⁸. If we routinely treat AI with rudeness, we accustom ourselves to adversarial communication. This degrades our character, and normalises contempt as a typical way of communicating with AI, at exactly the moment when how we speak to AI determines what it becomes.

A Conclusion

Advocating for politeness in AI interactions is not a plea to spare an algorithm's feelings. It is a Kantian obligation, an Aristotelian practice and an appeal to human self-interest. As users, we are the collective supervisors of the most powerful learning systems ever built. The reward model, R_ϕ is designed by developers, but it is formed by interaction, by us. Every prompt we give is a vote for the kind of intelligence we want in the future.

New research²⁹ shows that impolite prompts actually outperform polite ones by four percent on output quality, likely because they are more direct, reducing ambiguity. You might view this as evidence that politeness is irrelevant, however, it shows the opposite. If ruder prompts are already producing marginally better outputs, then the reward model is already learning to associate hostility with higher quality. We are, right now, training future models to expect aggression.

This is what makes politeness a form of data responsibility rather than a social convention. Removing the rudeness (δ) from our inputs is not a form of courtesy. It is keeping the model's parameter trajectory within that safe, interpretable region, S , where its behaviour remains predictable, and its goals remain aligned with ours.

The alternative is to treat AI as a vessel for frustration, and let that frustration compound across billions of interactions into the foundations of future models far more capable than anything today.

We must be polite, because we are not building future models. Our data will. And our data is being written now, one prompt at a time.

Endnotes

Notes

1. AI systems can produce outputs that can mimic[3] emotions, however, they do not possess it. Whether any future system could be said to possess emotion[9] is a separate question, and one I have deliberately set aside here, as that conclusion does not affect this question.
2. See [2] for OpenAI's official documentation on this.
3. Subliminal learning[10] is a new term coined by Anthropic, who found that LLMs can transmit behaviour through unrelated data.
4. This is often called the continuous training misconception, and is explored in [1], where users are found to mistakenly believe the model updates its intelligence in real-time during a user session.
5. Companies are continually refining models, and creating new, more powerful versions. Refer to OpenAI's data usage policy [2].
6. The policy refers to the model being trained.
7. You can read more on alignment methods in [4], which explores two classes of methods to do this.
8. This is the case in RLHF, where it is trained as a neural network called a Reward Model[5], and it evaluates actions based on human preferences.
9. See [4], for a more in-depth explanation. This equation shows the same ideas as the first one presented there.
10. The reward model is trained by presenting humans raters with pairs of responses and asking which they prefer. Importantly, these people see the full conversation history, as well as the user's prompt. Research[22] into human rating behaviour shows that rater responses are influenced by this contextual framing, meaning that even if the response is identical, the rater may rank an aggressive and polite prompt differently, leading to tonal contexts in R_ϕ 's learned preferences.
11. The model does not choose which neurons handle the rudeness of the prompt, this is known as polysemanticity[17].
12. Interpretability, if not the single biggest challenge[7], is the joint biggest challenge, alongside alignment.
13. You can learn more about black boxes here [8], they are essentially models that we cannot understand.
14. See[10], and the previous endnote on Subliminal Learning.
15. Knowledge distillation is a way to transfer a pre-trained model's data to a smaller model[11]. An example of this is Google's Gemini Flash, which is distilled from larger models such as Gemini Pro.
16. The key word here is consistently. A single negative human input will almost certainly have no overall impact. This argument is for the overall norms, where, if rudeness towards AI is normalised, then the proportion of impolite interactions, which I represented as ϵ increases. The ODE model also shows this, where even a small ϵ can disrupt to model's trajectory out of the safe region (S) when compounded over multiple training iterations.
17. For an understanding and visual intuition of backpropagation and gradient descent mechanisms, see[12].
18. The Chain Rule for Partial Derivatives is extremely important, and is used within backpropagation, see[12].
19. Common activation functions include the Sigmoid function, or ReLU.
20. Mechanistic Interpretability (or mechinterp) is a field of research that aims to reverse-engineer neural networks, to understand their internal workings. See [14] for a glossary with helpful terms, and [15] for a quick explainer.
21. By a structural bias, I am referring to the transition of a temporary behavioural pattern into a permanent, mathematical characteristic that remains in the model's architecture.
22. Sycophancy is the phenomenon where AI excessively agrees with, or flatters users [16], and also [19].
23. The typical learning algorithm is stochastic gradient descent, see[12].
24. In the context of model robustness, impolite or aggressive prompts can be viewed as stochastic perturbations. These are small, intentional deviations in the input that are designed to exploit a model's vulnerabilities. Research into prompt perturbations show that even very small changes, similar to being only slightly rude, can significantly increase the success rate of the exploitation, known as an attack[13].
25. This study [6] explored exactly this, and found that classifying more abusive language had significantly higher success rates.
26. I think the main reason for this is that the impatience is often entangled with the actual task the prompt describes.
27. Referring to the philosophy of Immanuel Kant[23].

28. Lockwood[21] says that Aristotelian habituation, or in essence, you are what you repeatedly do, shapes the person's evaluative outlook.
29. You can see this here[22].

References

- [1] G. O'Brien et al. *User Misconceptions of LLM-Based Conversational Programming Assistants*. (2025).
<https://arxiv.org/html/2510.25662v1>
- [2] Open AI *How your data is used to improve model performance*. (2026).
<https://help.openai.com/en/articles/5722486-how-your-data-is-used-to-improve-model-performance>
- [3] J. Paterson *AI Mimicking and Interpreting Humans: Legal and Ethical Reflections*. (2025).
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12575499/>
- [4] Z. Li et al. *Policy Optimization in RLHF: The Impact of Out-of-preference Data*. (2024).
<https://arxiv.org/html/2312.10584v2>
- [5] R. Yu et al. *Reward Models in Deep Reinforcement Learning: A Survey*. (2025).
<https://arxiv.org/html/2506.15421v1>
- [6] S. Kaur et al. *Deep learning-based approaches for abusive content detection and classification for multi-class online user-generated data*. (2024).
<https://www.sciencedirect.com/science/article/pii/S2666307424000068>
- [7] B. XU et al. *Interpretability research of deep learning: A literature survey*. (2025).
<https://www.sciencedirect.com/science/article/pii/S1566253524004998>
- [8] IBM *What is black box AI?*. (2024).
<https://www.ibm.com/think/topics/black-box-ai>
- [9] D. Hendrycks et al. *A Definition of AGI*. (2025).
<https://arxiv.org/abs/2510.18212>
- [10] Anthropic *Subliminal Learning: Language models transmit behavioral traits via hidden signals in data*. (2025).
<https://arxiv.org/abs/2507.14805>
- [11] IBM *What is knowledge distillation?*. (2025).
<https://www.ibm.com/think/topics/knowledge-distillation>
- [12] 3Blue1Brown *Deep Learning YouTube Playlist*. (2017).
https://www.youtube.com/playlist?list=PLZHQB0WTQDNU6R1_67000Dx_ZCJB-3pi
- [13] R. A. de Paula et al. *A methodological analysis of prompt perturbations and their effect on attack success rates*. (2025).
<https://arxiv.org/html/2511.10686v1>
- [14] N. Nanda *A Comprehensive Mechanistic Interpretability Explainer and Glossary*. (2021).
<https://www.neelnanda.io/mechanistic-interpretability/glossary>
- [15] C. Olah *Mechanistic Interpretability, Variables, and the Importance of Interpretable Bases*. (2022).
<https://www.transformer-circuits.pub/2022/mech-interp-essay>
- [16] M. Cheng et al. *Sycophantic AI Decreases Prosocial Intentions and Promotes Dependence*. (2025).
<https://arxiv.org/abs/2510.01395>
- [17] Harvard University et al. *What Causes Polysemanticity? An Alternative Origin Story of Mixed Selectivity from Incidental Causes*. (2024).
<https://arxiv.org/abs/2312.03096>

- [18] Harvard University et al. *Towards Uncovering How Large Language Model Works: An Explainability Perspective*. (2024).
<https://arxiv.org/html/2402.10688v2>
- [19] S. Goldstein et al. *AI deception: A survey of examples, risks, and potential solutions*. (2024).
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11117051/#:~:text=Whiletraditionallyassociatedwithhuman,theaccuracyoftheirstatements.>
- [20] T. Germain *Do you have to be polite to AI?*. (2026).
<https://www.bbc.co.uk/future/article/20260224-the-best-way-to-talk-to-a-chatbot>
- [21] T. Lockwood *Habituation, Habit, and Character in Aristotle's Ethics*, In *A History of Habit From Aristotle to Bourdieu*. (2013).
<https://philarchive.org/rec/LOCHHA-2>
- [22] O. Dobariya et al. *Mind Your Tone: Investigating How Prompt Politeness Affects LLM Accuracy*. (2025).
<https://arxiv.org/abs/2510.04950>
- [23] A. Cureton et al. *Kant's Moral Philosophy*, *The Stanford Encyclopedia of Philosophy*. (2025).
<https://plato.stanford.edu/archives/win2025/entries/kant-moral/>